

README FILE FOR THE SYNOPTIC SUMMARY OF THE DAY (SSOD) DATASET
Version 2.0.0

I. DOWNLOAD QUICK START

Bulk SSOD station data download access is available in three ways:

- 1) Period of record station files (data for all years and variables for a particular station)
- 2) Station/year files (one year of data for a particular station)
- 3) A tar file of all stations with data in a particular year

Data Access Locations	• HTTPS WAF: 1. <a href="https://www.ncei.noaa.gov/oa/synoptic-summary-of-the-day/v2/access/by-year/<YYYY>/SSOD_<station>_yyyy.csv">https://www.ncei.noaa.gov/oa/synoptic-summary-of-the-day/v2/access/by-year/<YYYY>/SSOD_<station>_yyyy.csv 2. <a href="https://www.ncei.noaa.gov/oa/synoptic-summary-of-the-day/v2/access/by-station/SSOD_<station>.csv">https://www.ncei.noaa.gov/oa/synoptic-summary-of-the-day/v2/access/by-station/SSOD_<station>.csv 3. <a href="https://www.ncei.noaa.gov/oa/synoptic-summary-of-the-day/v2/archive/<archived files>">https://www.ncei.noaa.gov/oa/synoptic-summary-of-the-day/v2/archive/<archived files> Where <station> refers to the GHCN identifier • Example file URL: 4. https://www.ncei.noaa.gov/oa/synoptic-summary-of-the-day/v2/access/by-year/2023/ssod_USW00003812_2023.csv
------------------------------	---

OVERVIEW

The Synoptic Summary of the Day (SSOD) version 2 dataset is a collection of Daily Climate Summaries derived from the synoptic and hourly reports provided in the Global Historical Climatology Network hourly (GHCNh) dataset. SSOD v2 is the successor to the Global Summary of the Day (GSOD) dataset, which was derived from the Integrated Surface Dataset or ISD. GHCNh is the successor dataset to ISD.

The SSOD daily summaries are based on hourly (e.g. METAR/FM15) or 6-hourly (SYNOP/FM16) reports for a calendar day that runs from 00 to 24 UTC. In this way, all daily climate summaries in SSOD nominally conform to the same synchronous 24-hour period (i.e., the same “synoptic day”) worldwide. This contrasts with the daily summaries provided in Global Historical Climatology Network daily (GHCNd) dataset. In [GHCNd](#), daily climate summaries are provided by national Met Services and are compiled according to the observation time and conventions used in their country for daily climate reports (e.g., midnight local time for US Airports, 06 UTC stations in Canada, 9 am local in Australia etc.).

The utility of SSOD is that daily summaries are provided in short time delay for the network of many thousands of stations that transmit hourly and synoptic reports. This ensures the SSOD

summary day is consistent for all stations worldwide, in theory making comparisons to numerical weather prediction model output and remotely sensed data more straightforward. One disadvantage of SSOD is that daily precipitation totals may be underreported in the synoptic messages. Additionally, maximum and minimum temperatures gleaned from the hourly or synoptic messages also may not correspond to the true daily extremes. The daily summaries provided by national Met Services in GHCNd are therefore recommended when the most accurate reports are required for daily rainfall and maximum/minimum temperatures. However, as noted above, the 24-hour periods represented in the daily summaries provided in GHCNd are not the same worldwide. This distinction between the consistent summary day in SSOD and the mix of summary days in GHCNd is the reason for the name change from Global Summary of the Day (GSOD) to Synoptic Summary of the Day (SSOD, version 2).

The headers used in csv files and definition of each daily element included in the dataset (as available from each station) are as follows:

Column Number		
1	STATION	GHCN identifier
2	Station_name	Station Name from source metadata
3	DATE	Date (YYYY-MM-DD) in UTC
4	Year	Year of Daily Summary (in UTC)
5	Month	Month of Daily Summary (in UTC)
6	Day	Day of Daily Summary (in UTC)
7	LATITUDE	Station Latitude (decimal degrees)
8	LONGITUDE	Station Longitude (decimal degrees)
9	ELEVATION	Elevation (meters)
10	mean_temperature	Average of synoptic hourly or synoptic + off synoptic hourly temperatures (from 00 to 24 UTC)--in °C

11	mean_temperature_Measurement_Code	Number of hourly values available (from 4 to 24). For a daily mean to be computed, at least one report in each of the 6-hour synoptic periods (00:00 to 05:59 UTC, 06:00 to 11:59 UTC, 12:00 to 17:59 UTC, and 18:00 to 23:59 UTC) must be available.
12	mean_dew_point_temperature	Average of synoptic hourly or synoptic + off-synoptic hourly dew point temperatures (from 00 to 24 UTC) in degrees C. For a daily mean to be computed, at least one report in each of the 6-hour synoptic periods (00:00 to 05:59 UTC, 06:00 to 11:59 UTC, 12:00 to 17:59 UTC, and 18:00 to 23:59 UTC) must be available.
13	mean_dew_point_temperature_Measurement_Code	Number of hourly values available (from 4 to 24)
14	mean_sea_level_pressure	Average of synoptic hourly or synoptic + off synoptic hourly dew point temperatures (from 00 to 24 UTC)--in hPa. For a daily mean to be computed, at least one report in each of the 6-hour synoptic periods (00:00 to 05:59 UTC, 06:00 to 11:59 UTC, 12:00 to 17:59 UTC, and 18:00 to 23:59 UTC) must be available.

15	mean_sea_level_pressure_Measurement_Code	Number of hourly values available (from 4 to 24)
16	mean_station_level_pressure	Average of synoptic hourly or syntopic + off synoptic hourly dew point temperatures (from 00 to 24 UTC)--in hPa. For a daily mean to be computed, at least one report in each of the 6-hour synoptic periods (00:00 to 05:59 UTC, 06:00 to 11:59 UTC, 12:00 to 17:59 UTC, and 18:00 to 23:59 UTC) must be available.
17	mean_station_level_pressure_Measurement_Code	Number of hourly values available (from 4 to 24)
18	mean_visibility	Average of synoptic hourly or syntopic + off synoptic hourly dew point temperatures (from 00 to 24 UTC)--in km. For a daily mean to be computed, at least one report in each of the 6-hour synoptic periods (00:00 to 05:59 UTC, 06:00 to 11:59 UTC, 12:00 to 17:59 UTC, and 18:00 to 23:59 UTC) must be available.
19	mean_visibility_Measurement_Code	Number of hourly values available (from 4 to 24)
20	mean_wind_speed	Average of synoptic hourly or syntopic + off synoptic hourly dew point temperatures (from 00 to 24 UTC)--in

		meters per second. For a daily mean to be computed, at least one report in each of the 6-hour synoptic periods (00:00 to 05:59 UTC, 06:00 to 11:59 UTC, 12:00 to 17:59 UTC, and 18:00 to 23:59 UTC) must be available.
21	mean_wind_speed_Measurement_Code	Number of hourly values available (from 4 to 24)
22	max_wind_speed	Maximum wind speed reported for the day—in meters per second
23	max_wind_gust	Maximum wind gust reported for the day—in meters per second
24	max_temperature	Maximum temperature for the day drive from hourly reports or from the climate code group in the SYNOP/FM12 report—in °C
25	max_temperature_Measurement_Code	* indicates value was derived from the hourly data (FM15/METAR/SA/ etc) rather than from the climate code group of the SYNOP/FM12
26	min_temperature	Minimum temperature for the day drive from hourly reports or from the climate code group in the SYNOP/FM12 report—For a daily mean to be computed, at least one report in each of the 6-hour synoptic periods (00:00 to 05:59 UTC,

		06:00 to 11:59 UTC, 12:00 to 17:59 UTC, and 18:00 to 23:59 UTC) must be available.
27	min_temperature_Measurement_Code	* indicates value was derived from the hourly data (METAR/SA/FM15 etc) rather than from the climate code group of the SYNOP/FM12
28	total_precipitation	Total precipitation reported for the day as reported in e.g., FM15/METAR hourly reports or SYNOP/FM12 reports –in mm
29	total_precipitation_Measurement_Code	n (Number of hours) = Sum of hourly reports, where n is the count of hourly totals reported A = 1 report of 6-hour precipitation amount. B = Summation of 2 reports of 6-hour precipitation amount. C = Summation of 3 reports of 6-hour precipitation amount. D = Summation of 4 reports of 6-hour precipitation amount. E = 1 report of 12-hour precipitation amount. F = Summation of 2 reports of 12-hour precipitation amount. G = 1 report of 24-hour precipitation amount. H = Station reported '0' as the amount for the day (eg, from hourly, 6-hourly reports, etc.) or the amount could not be

		calculated, but at least one occurrence of precipitation was noted in the present weather observations. This could indicate that a trace occurred (or more if the value is missing). I = Station did not report any precipitation data for the day and did not report any occurrences of precipitation in its present weather observations.
30	snow_depth	From the last report for the day if reported more than once—in mm
31	observed_pres_wx	Observed weather reported across all available reports for the day for weather types “Fog/Rain/Snow or Ice Pellets/Hail/Thunder/Tornado or Funnel Cloud” The presence or absence of the above weather types is noted as a zero ('0') or a one ('1'): reported=1; no report=0 yielding a string of six digits (one digit for each weather type).
